INCH-POUND

MIL-R-70560B (AR)
04 December 1991
SUPERSEDING
(See 6.6)

MILITARY SPECIFICATION

RADAR SET AN/VPS-2A: 9360797
(RANGE ONLY RADAR)

This specification is approved for use by the U. S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and tests for a Radar Set AN/VPS-2A, hereafter referred to as the radar. The radar is used in the Gun, Anti-Aircraft, Artillery, Self-Propelled 20MM, M163A2, and Towed 20MM, M167A2. The radar is a coherent, X-band radar, which measures target range and uses the doppler principle to detect the target and remove clutter interference. The radar set measures target distance along the line of sight as range and continuously supplies this data to the fire control processor (FCP). The radar is comprised of six major units. The antenna (Unit 1) consists of the reflection and feed assembly, traverse and elevation servo drive assemblies, and waveguide traverse and elevation rotary joints. The receiver-transmitter (Unit 2) generates the high power transmit pulse for radiation by the antenna and picks up the target echoes and sends them to the receiver. The receiver (Unit 3) receives the target echoes from the receiver-transmitter and sends them to the range computer. The range computer (Unit 4) generates range and range rate digital data from the receiver signals and supplies this information to the fire control processor where it is used in computing target motion lead angles. The power supply (Unit 5) provides, distributes and regulates all power required by the radar. The distribution box (Unit 6) selects the mode of operation for antenna positioning and serves as a junction point for electrical interface between the antenna and other units of the radar set. Unit 2, Unit 4 and Unit 5 each contain Built-In-Test (BIT) circuitry to monitor signals and provide fault isolation capability (see 6.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 1285

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926 -	Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-I-45607 -	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-R-50674 -	Receiver, Radar R-1475/VPS-2: 10548338 (Unit 3)
MIL-A-50687 -	Antenna Assembly, AS-2049A/VPS-2: 12563312 and 12563321
MIL-R-70561 -	Receiver-Transmitter, Radar RT-860A/VPS-2A: 9361152 (Unit 2)
MIL-C-70562 -	Computer, Range CP-888A/VPS-2A: 9361153 (Unit 4)
MIL-P-70563 -	Power Supply: PP-4812A/VPS-2A: 9361154 (Unit 5)
MIL-D-70704 -	Distribution Box: 9397370 (Unit 6)

STANDARDS

MILITARY

MIL-STD-109 -	Quality Assurance Terms and Definitions
MIL-STD-810 -	Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those cited in the solicitation.

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DRAWINGS (See 6.4)

U. S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9360797 - Radar Set AN/VPS-2A
9361217 - TS-3211/TSM - 100B, Remote Bit Test Set
9361225 - TS-3211/TSM - 100B, Remote Bit Test Cable
9397549 - Radar Set AN/VPS-2A Interconnecting Wiring Diagram
10548164 - IF Amplifier
11748124 - Radar System Tester AN/VPM-2
12950877 - Requirements for Soldering Electrical and Electronic Assemblied

SPECIAL PACKAGING INSTRUCTION

PS 9360797 - Packaging of Radar Set, AN/VPS-2A

(Copies of drawings, publications and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9360797, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the radar shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.
b. Inorganic protective surface finishings.
c. Part identification and marking.
d. Workmanship.

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3.2.1 Subassemblies. The units of the radar shall have been inspected and accepted in accordance with their respective drawings and specifications prior to their installation in the radar. The following units are covered by the indicated specifications:

UNIT NO.	NAME	SPECIFICATION
1	Antenna Assembly AS-2049A/VPS-2A: 12563321	MIL-A-70708
2	Receiver-Transmitter, Radar RT-860A/VPS-2A: 9361152	MIL-R-70561
3	Receiver, Radar R-1475/VPS-2A: 10548338	MIL-R-50674
4	Computer, Range CP-888A/VPS-2A: 9361153	MIL-C-70562
5	Power Supply PP-4812A/VPS-2A: 9361154	MIL-P-70563
6	Distribution Box: 9397370	MIL-D-70704

3.3 Environmental service conditions.

3.3.1 Vibration. The radar shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to the equivalent of 6000 miles of track-laying vehicle operational vibration as defined by MIL-STD-810 narrowband random-on-random data for equipment installed on the deck of the M113 armored personnel carrier.

3.3.2 Shock. The radar shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude operating. The radar set shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

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3.3.4 Temperature.

3.3.4.1 Operating. The radar shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40 deg F and +155 deg F.

3.3.4.2 Non-operating (storage). The radar shall not suffer permanent damage when subjected to the constant temperature extremes of -60 deg F for a period of 24 hours and +160 deg F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The radar shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the radar.

3.5 Performance. The radar shall meet the requirements of this section when supplied with 22 volts direct current (Vdc) to 30 Vdc from a 750 watt (W) source and 115 V $\pm$ 5 percent, 400 Hz $\pm$ 1 percent, from a 15 W source.

3.5.1 Standby power. The radar shall use the 22 Vdc to 30 Vdc primary power to activate its standby circuits. The blowers in Units 2 and 5 shall be activated upon application of standby power which shall enable the radiate function 120 $\pm$ 12 seconds after application of standby power.

3.5.2 Radar-ready signal. The radar shall provide a single-pole double-throw (SPDT) switch action to the FCP, indicating that the radar is ready to radiate.

3.5.3 Radiate function. The radiate function shall be available through closure of 2S6 (SERV RAD CONT) in the receiver-transmitter unit cover or through application of +22 Vdc to +30 Vdc at 2J1 pin R from a remote control unit.

3.5.4 Unit 2 interlock. The Unit 2 interlock switch shall close when the cover of the RT unit is tightened down enabling Unit 2 operation. When the RT unit cover is open, the interlock switch shall open, preventing operation of Unit 2. The manual override feature shall permit operation of Unit 2 when the RT unit cover is open and the interlock switch is in the manual override position.

3.5.5 Transmit characteristics.

3.5.5.1 Average power. The minimum average transmit power on any channel shall be 4.26 W at 2J3.

3.5.5.2 RF transmit mode.

a. The radar transmit pulse shape shall be cosine squared

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(COS^2).

b. The radar transmit pulse width shall be 280 $\pm \frac{20}{10}$ nanoseconds (ns) at 50 percent points.

3.5.5.3 Transmit frequency. The receiver-transmitter unit shall be capable of operation on six discrete frequencies by selecting one of six oscillator plug-in modules. The frequency of the transmitted signal shall be 9205 megahertz (MHz) ± 10 MHz using channel 1 oscillator module. Using the channel 1 measurement as a reference, the channel 2 through 6 output frequency shall be as follows:

Channel 2	Reference +8 ± 2 MHz
Channel 3	Reference +16 ± 2 MHz
Channel 4	Reference +24 ± 2 MHz
Channel 5	Reference +32 ± 2 MHz
Channel 6	Reference +40 ± 2 MHz

3.5.5.4 Pulse repetition frequency (PRF). The PRF shall be not less than 19,978 and not more than 19,979 pulses per second.

3.5.6 Midrange cal function. The midrange cal function shall provide a self test of the radar ranging circuits. When midrange cal switch (4S1) on the range computer is depressed, midrange cal lamp (4DS1) shall illuminate brightly.

3.5.7 System noise figure. The system noise figure shall be 13 dB maximum at room temperature (70 deg F).

3.5.8 Receiving characteristics.

3.5.8.1 Clutter lock-on function. The clutter lock-on function shall enable radar lock-on of the stationary targets for test purposes when the CLUTTER LOCK-ON switch on Unit 2 is in the test position.

3.5.8.2 Range and mode data. The accuracy of the range output shall be ± 10M for targets of 1000 meters range or less and ± 1% for targets at greater than 1000 meters to the maximum range of 5000 meters.

3.5.8.3 Subclutter visibility. The subclutter visibility shall be a minimum of 40 DB with or without automatic gain control (AGC) under MDS conditions. The subclutter visibility with AGC at a clutter level of -15 dBm shall be 40 dB minimum.

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3.5.8.4 AGC characteristics. With a clutter signal input the AGC circuit shall maintain a constant output from the intermediate frequency (i-f) amplifier from the point of AGC cut-in to 0 dBm input signal at 2J3 with a flatness of ± 1 dB. Under these same conditions, the sum channel noise level at 4TP46 and 4TP47 shall not increase more than 1 dB. The AGC circuit shall be stable in the presence of clutter signals of 0 dBm level at 2J3.

3.5.8.5 Active ECM indication. The radar shall provide a bit in the digital range word to activate the remote panel ECM indication when the signal level is 20 ± 5 dB greater than the lock threshold and the target velocity is less than 10 M/s. The ECM detect signal shall cause the TRACK/JAM/RADIATE lamp on the M61A1 sight to flash at a 1 Hz rate.

3.5.8.6 Three-second fade logic. When the velocity of the target falls below 10 M/s, the radar shall maintain the track mode for 3.00 ± 0.25 seconds after loss of doppler signal.

3.5.8.7 One-second fade logic. When the radar is tracking a target with a velocity between 10 M/s and 310 M/s and target return signal reduces to a value below the minimum discernible signal (MDS) of the radar, the radar shall maintain the track mode for 1.0 ± 0.02 seconds after the signal level drops below the MDS of the radar with the range rate equal to the last value of range rate before loss of signal ± 20 M/s.

3.5.8.8 Range rate lag. The range rate lag, when tracking a target which is accelerating from 0 to 243 M/s at an 81 M/s^2 rate, shall not exceed 30 M/s.

3.5.8.9 Minimum range tracking. The radar shall track an inbound target to a minimum range of 250 meters before breaking track.

3.5.8.10 Receiver bandwidth. The 3 dB bandwidth of the radar receiver shall be 1.9 ± 0.2 MHz.

3.5.9 Antenna characteristics. The antenna shall meet all the requirements of MIL-A-70708.

3.5.10 BIT assemblies. Radar Units 2, 4 and 5 shall contain BIT assemblies that monitor radar performance and provide range data to an external fire control processor. The performance of the BIT assemblies shall be as specified in MIL-R-70561, MIL-C-70562 and MIL-P-70563, respectively.

3.6 First article. When specified in the contract or purchase order, sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

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3.7 Soldering. All soldering shall be in accordance with the general requirements of Note 2 of Drawing 12950877.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5)

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+60°F to +90°F (+16°C to +32°C)
b.	Relative humidity	90% RH max.
c.	Pressure	28 to 32 inches of Hg (700 to 815 mmHg)

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4.3 First article inspection.

4.3.1 Sample. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

Name	Drawing	Quantity
Radar Set	9360797	3

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogenous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies.

a. The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawings and specifications utilizing as a minimum the conformance criteria specified in this document and the

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Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

b. Alternative inspection procedures, methods or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the procuring contracting officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program plan or detailed inspection system, as applicable.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in Table I shall be performed on a 100% basis.

TABLE I. Functional tests (100 percent inspection).

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
101. Electrical	3.4	4.7.2
102. Performance	3.5	4.7.3
Minor:	None	N/A

4.5 Special sampling. One radar out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table I and shall then meet the tests in Table II. Unless otherwise

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specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE II. Special sampling.

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
103. Vibration	3.3.1	4.7.1.1
104. Shock	3.3.2	4.7.1.2
105. Altitude	3.3.3	4.7.1.3
106. Temperature	3.3.4	4.7.1.4
107. Humidity	3.3.5	4.7.1.5
Minor:	None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials,

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dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table III (Test equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

TABLE III. Test equipment list.

ITEM NUMBER	DESCRIPTION
1. Audio oscillator	Frequency: 600 Hz to 20 KHz.
2. Power supply	0 to 36 V @ 30 A, regulation ± 0.005 percent, ripple less than 1 mV rms.

TABLE III. Test equipment list. (continued)

ITEM NUMBER	DESCRIPTION
3. Oscilloscope	Dual trace, 150 MHz bandwidth, both channels are identical. Rise time: 2.4 ns with or without probe. Vertical deflection: 5 mV/div. dc to 60 MHz, rise time: 5.9 ns; 10 mV/div. dc to 100 MHz, 3.5 ns rise time. Calibrated deflection: From 5 mV/div. to 10 V/div. in 11 steps. Input RC: 1 megohm paralleled by 20 pF. Horizontal deflection: 0.05 microsec/div. to 5 s/div. in 25 steps. Triggering: Dc, to 150 MHz, x10 magnifier. Calibrated sweep delay: 1 microsec to 50 sec. Sweep accuracy: ± 3 percent.

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TABLE III. Test equipment list. (continued)

ITEM NUMBER	DESCRIPTION
4. Waveguide RF load	Frequency range: 8.2 to 12.46 Hz. UG-39/U flange. VSWR: Max. of 1.10 to 1. Maximum average power: 500 W, peak power not less than 3 kW.
5. Digital voltmeter	Dc voltage ranges 1, 10, 100, 1000 with 20 percent over-range on all ranges. Automatic polarity. Resolution of 0.01 percent of range (100 microV on 1 V range max.). Accuracy: ± 0.02 percent of input, +0.01 percent of range. Dc input resistance: Constant 10 megohms on all ranges. Overload protection: ± 1200 Vdc or ± 1700 V peak ac indefinitely on any range without damage.
6. RF power meter	Automatic self balancing, 7 calibrated ranges from 10 microW to 10 MW and -20 to +10 dBm full scale in 5 dB steps; accuracy of ±1 percent of full scale, external compensated thermistor mount required.
7. Thermistor mount	Frequency range: 10 MHz to 18 GHz. Maximum SWR: 1.35 to 1 from 1 to 12.4 GHz. Operating impedance: 200 ohms, 50 ohm match over frequency range. Temperature compensated 0.15 MW maximum input.
8. Waveguide frequency meter	Frequency range: 8.2 to 12.4 GHz. UG-39/U flanges. Overall accuracy: 0.08 percent. Dial increments: 5 MHz.
9. Waveguide coupler	Frequency range: 8.2 to 12.4 GHz. UG-39/U flanges. Coupling: 40 ± 0.3 dB. Coupling sensitivity: ± 0.5 dB max. Minimum directivity: 40 dB. Main line VSWR: 1.10 to 1 max. Secondary line VSWR: 1.20 to 1 max.

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TABLE III. Test equipment list. (continued)

ITEM NUMBER	DESCRIPTION
10. True rms voltmeter	Range: 1 mV to 300 V, 12 ranges. dB range: -72 to +52 DBM (600 ohms). Frequency range: 10 Hz to 10 MHz. Meter accuracy: ± 1 percent from 600 Hz to 10 KHz. Input impedance: 0.001 V to 0.3 V ranges, 10 megohms shunted by less than 50 pF, 1.0 V to 300 V ranges 10 megohms shunted by less than 20 pF. Overload protection: 30 dB over full scale.
11. X-band argon tube noise source	Frequency range: 8.2 to 12.4 GHz. Excess noise ratio: 15.75 ± 0.5 dB. Anode current: 175 mA. UG-39/A connector. Unfired SWR: 1.3 to 1; Fired SWR: 1.2 to 1. Maximum unfired insertion loss: 0.5 dB. Maximum fired insertion loss: 20 dB. Termination power rating: 4 W.
12. Noise source power supply	Adjustable output current: 0 to 250 mA. Separate controls for set and fire of noise source. Output current meter range: 0 to 250 mA.
13. Waveguide variable attenuator	Frequency range: 8.2 to 12.4 GHz. UG-39/U flanges. Attenuation range: 0.5 to 60.0 dB. Accuracy: ± 2 percent or ± 0.1 dB of reading whichever is greater from 0 to 50 dB and ± 3 percent, 50 to 60 dB. VSWR: Less than 1.15 to 1. Average power capacity rating: 10 W.
14. RF variable attenuator	Attenuation 101 dB in 1 dB steps ± 0.3 dB over entire range. Frequency range: Dc to 1 GHz. Power, average: 1 W. Impedance: 50 ohms.

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TABLE III. Test equipment list. (continued)

ITEM NUMBER	DESCRIPTION
15. Dynamic range simulator	Target range: 0-999,999 ft with incremental resolution of 1 ft, accuracy of ± 1 ft. Range jitter: Less than 2 ns +0.001 percent of range set. Target velocity: 0.0-99,999.9 ft/sec. with incremental resolution of 0.1 ft/sec. accuracy of ± 0.005 percent of instantaneous velocity. Target acceleration: 0.000-9,999.99 ft/sec^2 with incremental resolution of 0.01 ft/sec^2 accuracy of ± 0.005 percent of set acceleration. Reversals: Upon reaching zero range on any dynamic run, velocity direction automatically reverses; upon reaching zero velocity on any dynamic run, both velocity and acceleration automatically reverse. Trigger PRF: Internal or external. Start and stop settings for range run: Direction of range, change switch, reset switch, start-run switch, target fade switch. Target pulse output: 6 V min. into 50 Kohm equal to or less than 10 ns rise time, 400 ns width, positive polarity. External trigger input: 7 V peak max. Rep rate: 40 KHz max, 0.33 Kohm impedance, positive polarity.
16. Phase shifter	Center frequency: 60 MHz. Bandwidth: 10 percent. Phase shift range: 0 to 180 degrees. Impedance: 50 ohms. Insertion loss: 1 dB max. Manual control of phase shift.

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TABLE III. Test equipment list. (continued)

ITEM NUMBER	DESCRIPTION
17. Dual channel recorder	Variable paper speeds: 1, 5, 20 and 100 mm/s. Frequency response: Dc to 125 Hz. Range selection: V/cm or MV/mm of 0.5, 1, 2, 5, 10 or 20. Variable gain knob. Marker capability. NOTE: All specifications apply to each channel.
18. X-band fixed attenuator	6 ± 0.3 dB over entire frequency range. Frequency range: Dc to 12.4 GHz. Power capability: (Max.) average 2 W, peak, 0.3 KW at duty cycle 0.001. Impedance: 50 ohms. VSWR: No worse than 1.3 to 1 across the band.
19. RF detector	Frequency range: 10 MHz to 12.4 GHz. Frequency response: ± 0.5 dB over full range. Maximum input power: 100 mW. Sensitivity with no load: Input no greater than 0.4 mW to give 0.1 V rectified output. Square law response: Within ±0.5 dB from low level up to 1 mW input. VSWR: Less than 1.4 to 1 to 10 GHz; less than 1.5 to 1 from 10 to 12.4 GHz. Output capacitance: 30 microF. Video internal impedance: At 0 dBm input, 5 Kohms max; at -30 dBm input 20 Kohms max.
20. Precision dc power supply	Dc from 0 to 25 V at 500 mA, 4 decade setability and readout. Accuracy: ± 0.1 percent with 1 mV vernier. Stability: 100 microV for 8 hours. Resolution: 10 microV. Peak-to-peak ripple noise: 100 microV. Source impedance: 0.20 milliohms. Adjustable current limiting meter for V or mA monitoring. Regulation: 0.0005 percent plus

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500 microV.

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TABLE III. Test equipment list. (continued)

ITEM NUMBER	DESCRIPTION
21. Frequency counter	Dc coupled 0 to 50 MHz. Gate time: 1 micros to 10 s. Accuracy: ± 1 count ± time base accuracy. Readout: In KHz or MHz with positioned decimal point display, 8 digits in line. Adjustable trigger level. Sensitivity: 100 mV rms. Sensitivity switch with ranges of 0.1, 1 and 10 V.
22. RF signal generator	Frequency range: 50 KHz to 65 MHz. Accuracy: ± 1 percent, fine frequency vernier. Output level: Continuously adjustable from 0.1 microV to 3 V into 50 ohms. Output attenuator: Calibrated in 10 dB steps from 3 V full scale to 1.0 microV full scale, and also calibrated in dBm, with vernier control for continuous adjustment between 10 dB steps. RF output meter: Calibrated in dBm and volts. Auxiliary fixed level CW output jack minimum of 200 mV rms into 50 ohms from 19 to 65 MHz.
23. RF voltmeter	Frequency range: 20 KHz to 1.2 GHz. Accuracy: ± 3 percent. Voltage sensitivity: 1 mV. dB range: 81 dB. Input impedance: Greater than 100 Kohm. VSWR not more than 1.2 to 1; must include 50 ohm termination matched to meter and/or probe.
24. Multimeter	Dc voltage ranges: 2.5, 10, 50 V. Polarity reversal switch. Accuracy: ± 2 percent FS.
25. RF test set	AN/VPM-2 Radar System Tester 11748124, or equivalent.
26. System control unit	As specified in Figure 1. See Table IV for fixture parts list.

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TABLE III. Test equipment list. (continued)

ITEM NUMBER	DESCRIPTION
27. Headset, H-140 AU	NSN 596500-892-1010.
28. Remote BIT Test Set	TS-3211/TSM-100B, 9361217
29. RBT cable	TS-3211/TSM-100B, 9361225,

TABLE IV. System control unit parts list.

ITEM	DESCRIPTION
F1	Fuseholder and 1 A fuse
P1	Connector, MS3106B16-105
P2	Connector, L22TF21S6N-A
K1	Relay, DPDT, 24 Vdc coil
S1, S3	Switch, SPST, 2 Adc
S2	Switch, DPST, 5 Adc
DS1-DS3	Lamps, 24 Vdc
DS4	Lamp, 115 V
M1	Meter, running time: 0 to 1000 hours

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, Procedure I, of MIL-STD-810 and paragraph 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph II-2.1 apply. The radar shall be attached to the vibration generator directly or with a fixture and securely held by its normal means of attachment. After vibration, the radar shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

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4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz.

After the application of all the shock impulses, the radar shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating). This test shall be performed as specified in Procedures II, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

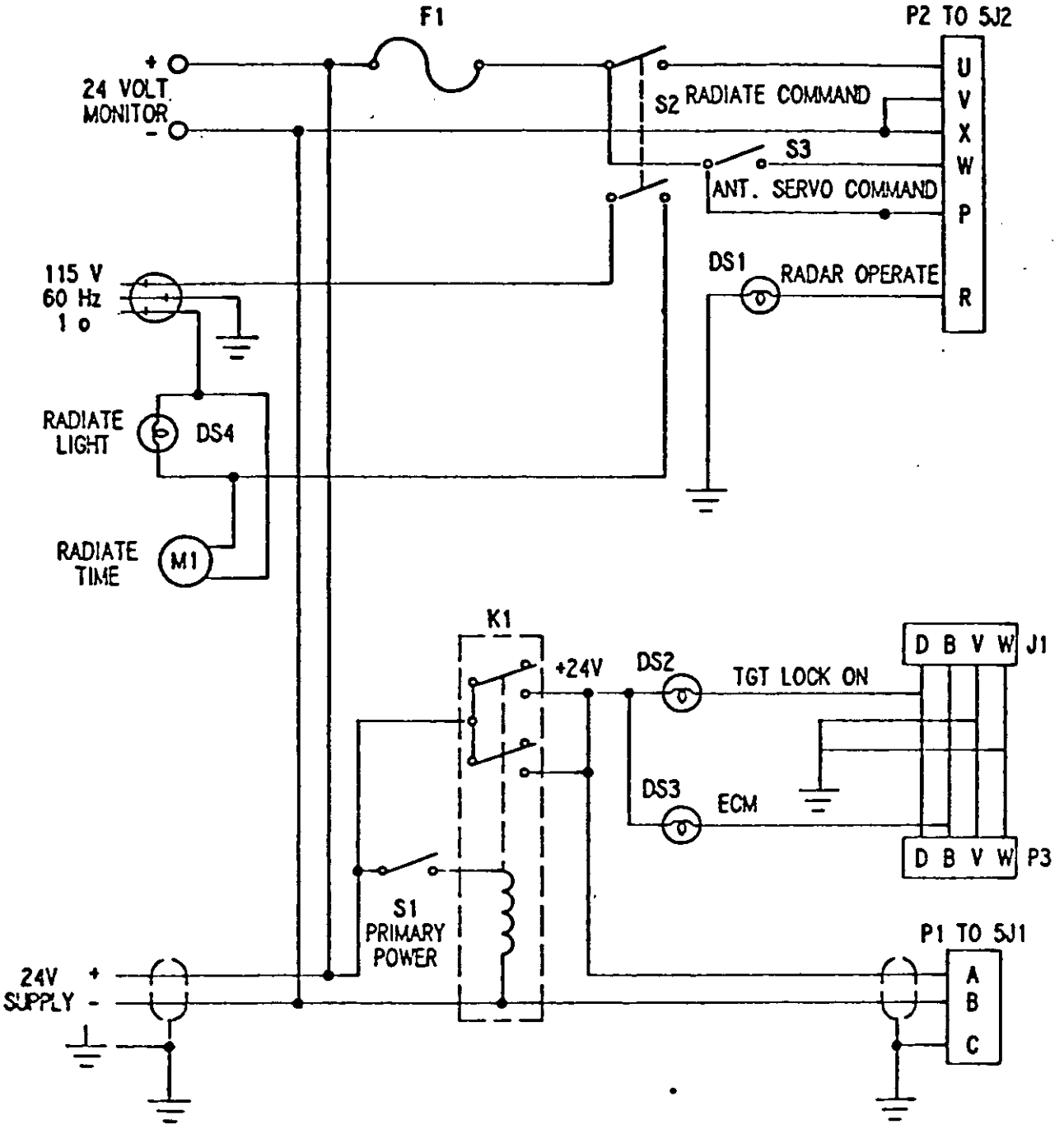
4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

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NOTE: ALL OTHER PINS ON J1 AND P3 ARE WIRED ONE TO ONE. P3 TO 4J1. J1 TO W103 P3.

FIGURE 1. System control unit diagram.

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4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

4.7.3 Performance. The selected radars shall be tested as follows:

4.7.3.1 Initial test setup. Connect the radar and test equipment as shown in Figure 2 with the controls of Units 2 and 4 of the units under test and the system control unit (item 25, Table III) set as follows:

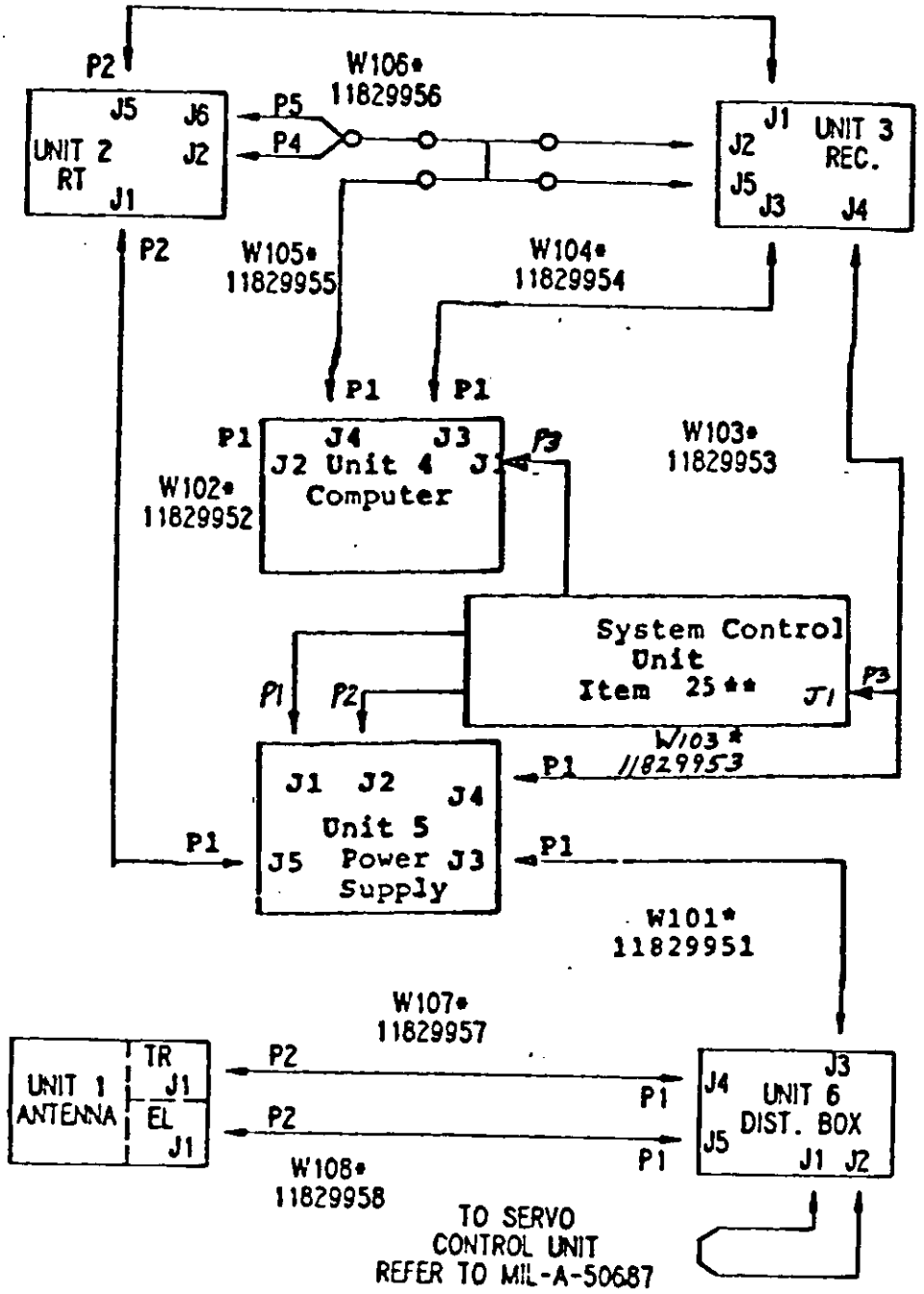
Unit	Units Under Test Designation	Position
2	S-2 Clutter Lock On Switch	Normal
2	S-3 Door Interlock	Pull Out
2	S-6 Ser Radiate Control	Off
4	S-2 Mode	Normal
4	S-3 Accel	Pos.
4	S-4 Return	TGT
4	S-5 Range	Run
4	S-6 Rate	Run
5	CB1 - Converter Circuit Breaker	Depressed
5	CB2 - Traverse Circuit Breaker	Open
5	CB3 - Elevation Circuit Breaker	Open
5	CB4 - Control Circuit Breaker	Depressed
5	CB5 - Blowers	Depressed
5	CB6 - 6.5 Vdc	Depressed

SYSTEM CONTROL UNIT

DESIGNATION	
S1 - PRIMARY POWER	OFF
S2 - RADIATE COMMAND	OFF
S3 - ANT. SERVO COMMAND	OFF

With the digital voltmeter (VM) connected to +24 Vdc MON jacks on the system control unit (item 25, Table III) adjust the power supply (item 1, Table III) for +28.0 $\pm$ 0.1 Vdc.

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* CABLE NUMBERS LISTED TO PROVIDE WIRING INFORMATION FOR SYSTEM TEST CABLES.
** REFER TO 4.6.2.3

FIGURE 2. System interconnecting cabling.

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4.7.3.2 Standby power. With switch S1 PRIMARY POWER on the system control unit on, note time of the turn-on in order to monitor the time delay. In the meantime, verify that the fan in Unit 2 and the fan in Unit 5 are operating. Continue time monitor and note when the RADAR OPERATE light (+28 V between RJ2-R(+) and E1(-) Unit 5) on the system control unit lights. The total delay time shall be 120 $\pm$ 12 seconds.

4.7.3.3 Unit 2 interlock check. With the lid on Unit 2 box closed and screwed down, and the RADIATE command of the system control unit on, the power meter (item 5, Table III) shall read a minimum of 4.26 W. Turn the RADIATE command switch OFF and open the cover of Unit 2.

4.7.3.4 Transmit characteristics.

4.7.3.4.1 Transmit average power out. With test equipment set as shown in Figure 3 and the SERV RADIATE CONTROL of Unit 2 of the radar on, the RF power meter shall read a minimum of 4.26 W.

4.7.3.4.2 Transmit pulse width. With the 6 dB attenuator (item 17, Table III) and the X-band detector (item 18, Table III) connected to the 40 dB coupler and a cable from the detector to the oscilloscope terminated in 50 ohms, the detected output pulse must be 280 $^{+20}_{-10}$ ns measured at the 50 percent amplitude points.

4.7.3.4.3 Transmit frequency. The frequency as read on the waveguide frequency meter (item 7, Table III) must meet the requirements of 3.5.5.3.

4.7.3.4.4 Radar PRF. The PRF of the radar as read on the frequency counter (item 20, Table III) must be 19.978 +0.001, -0.000 KHz.

4.7.3.4.5 Repeat 4.7.3.4.1 through 4.7.3.4.4 for the five remaining channels.

4.7.3.5 Midrange cal function.

4.7.3.5.1 Lock threshold setting.

a. Remove attenuator 3AT5 and replace with RF attenuator (item 13, Table III).

b. Set the RF attenuator for 2 dB attenuation.

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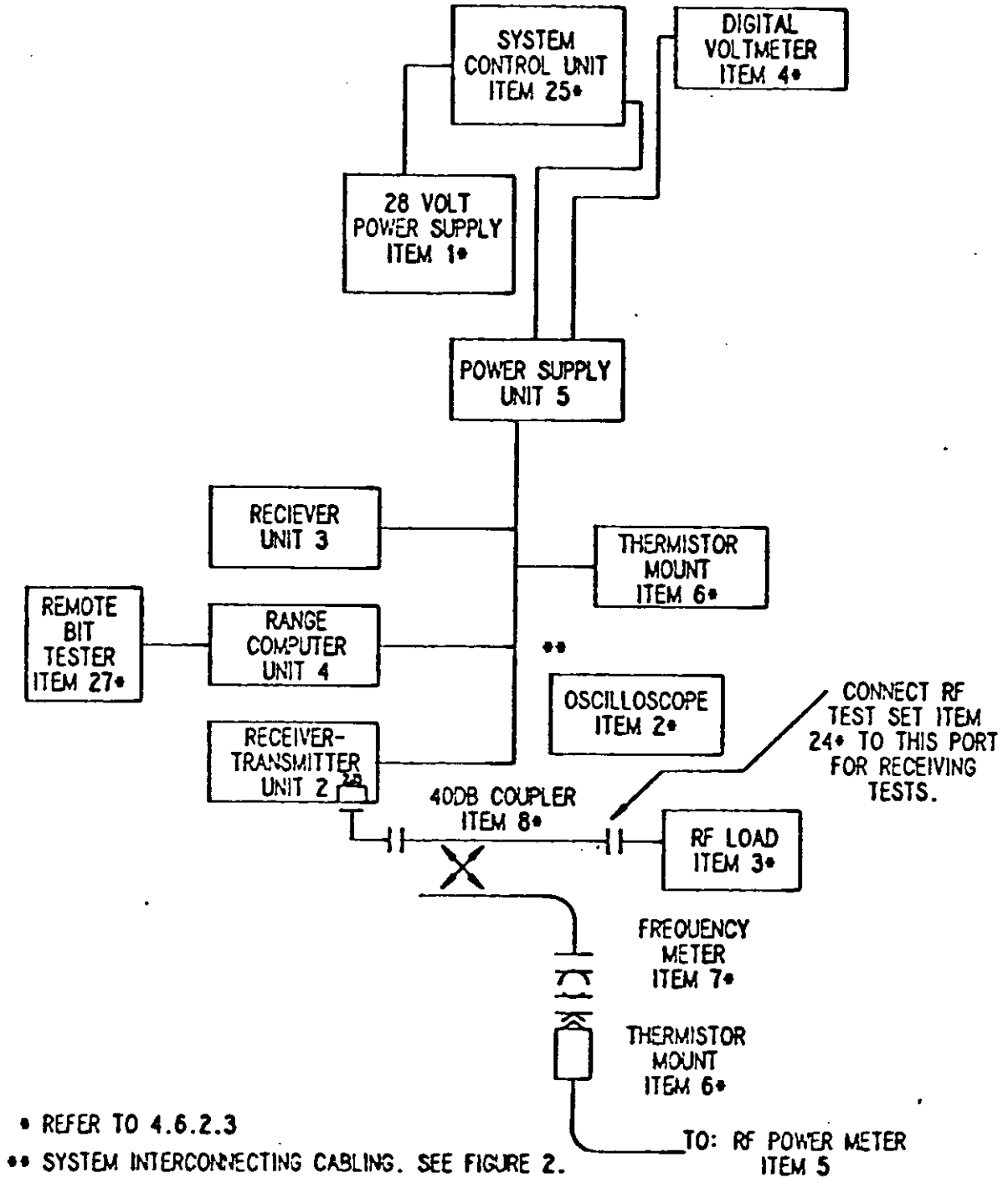


FIGURE 3. Transmitting and receiving characteristics test setup.

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c. Adjust 4A15R22 ccw until the TARGET LOCK-ON light on the system control unit comes on.

d. Adjust 4A15R22 ccw until the TARGET LOCK-ON light just goes out without flicker.

e. Remove the variable attenuator and reinstall 3AT5.

4.7.3.5.2 Midrange cal test. With the radar in the standby mode, depress the midrange calibrate button on Unit 4 of the radar and observe that the midrange cal light illuminates brightly.

4.7.3.6 System noise figure. With the test equipment connected as shown in Figure 4, the variable attenuator (item 12, Table III) set for maximum attenuation, the variable attenuator (item 13, Table III) set for 0 dB, the RANGE switch of Unit 4 set to FREEZE, and the noise tube current set for 175 mA, note the noise voltage level on item 10. With variable attenuator (item 13, Table III) set for 3 dB, reduce variable attenuator (item 12, Table III) until noise voltage level is the same as noted above. Difference between variabe attenuator reading and the excess noise ratio of argon tube noise source must be a noise figure of 13 dB maximum.

4.7.3.7 Receiving characteristics. The following self-test procedure shall be acceptably performed on RF test set (item 24, Table III) before using it for receiver characteristics measurements.

4.7.3.7.1 AN/VPM-2 self-test procedure.

a. Voltage checks.

1. Connect the equipment as shown in Figure 5. Do not connect flexible cable W5 at this time.

NOTE

Any time the AN/VPM-2 system tester power is interrupted for more than 15 minutes, voltage checks and self-checks should be repeated to ensure that the equipment is fully operational before continuing any tests.

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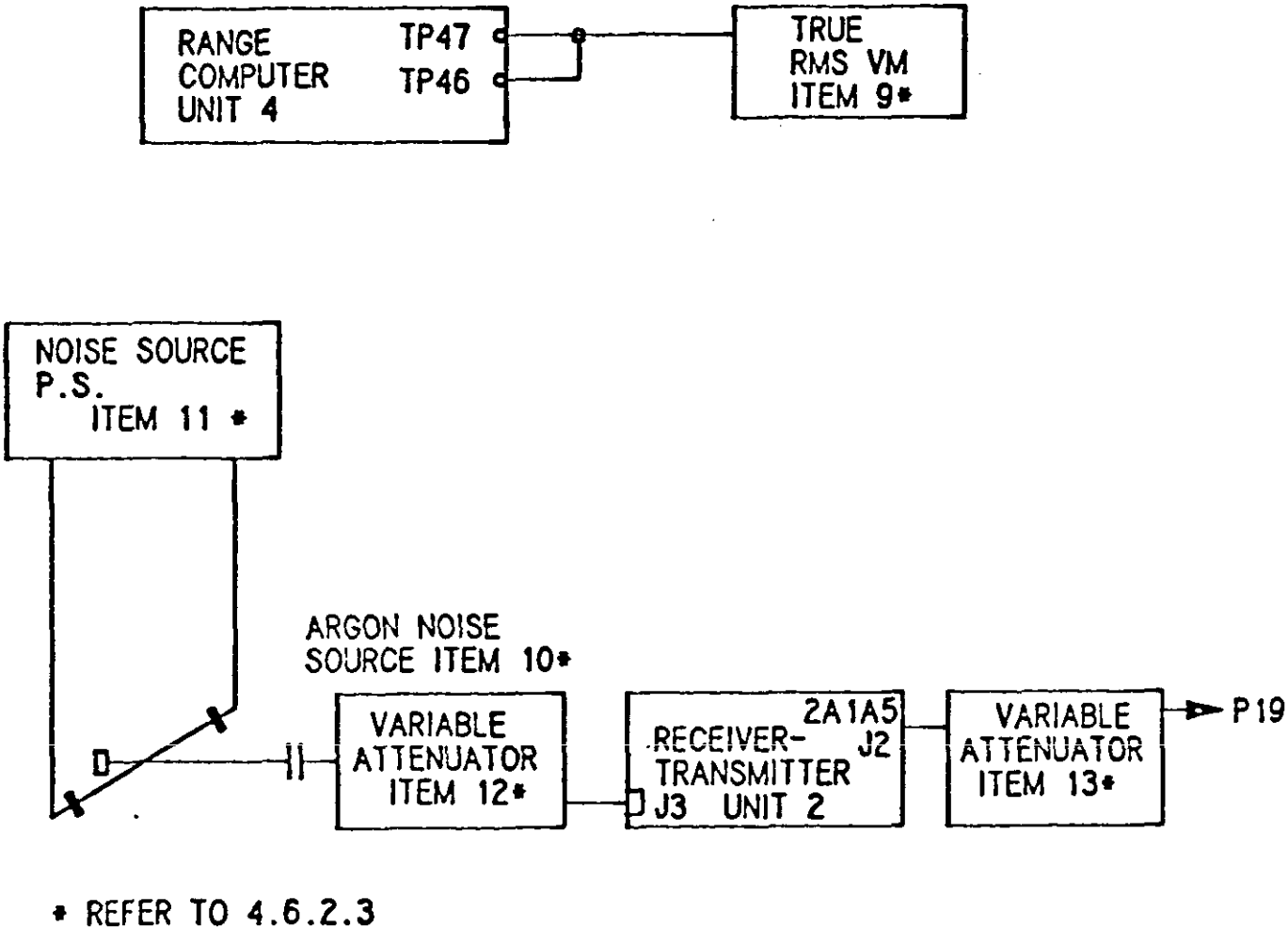


FIGURE 4. Receiver noise test setup.

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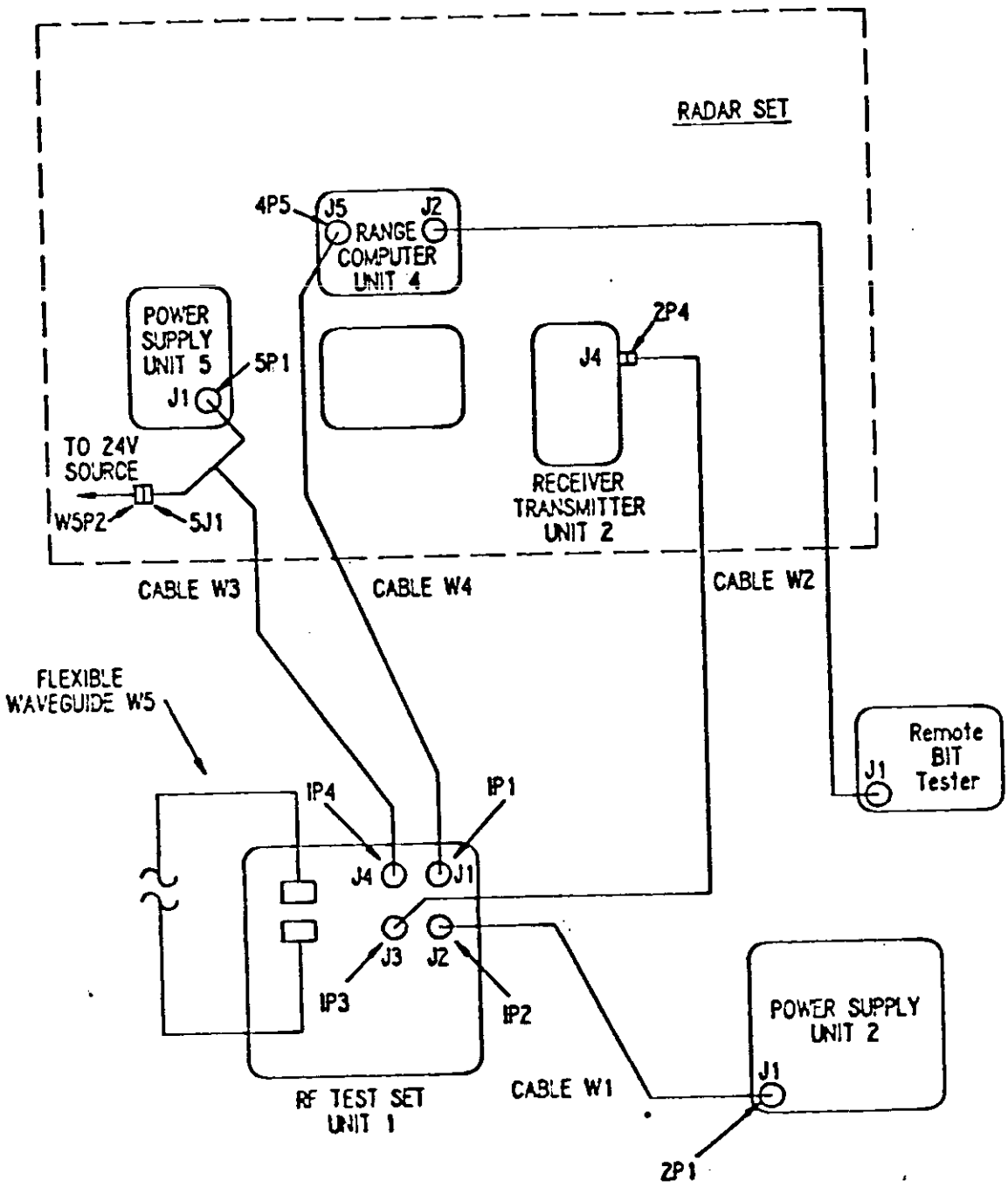


FIGURE 5. AN/VPM-2 system tester self-test, equipment setup.

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2. Set the AN/VPM-2 system tester controls as follows:

Control	Position
POWER	OFF
FUNCTION SELECT	VOLTAGE
VOLTAGE SELECT	+24
RF LEVEL	20 dB
FREQ SELECT	Radar transmit frequency (see Table V)
MODE SELECT	SELF TEST
PHASE CONTROL	MIDRANGE
CLUTTER	OFF
CLUTTER LEVEL	20 dB
RANGE SELECT	R3
TARGET LEVEL	27 dB
TARGET FREEZE	OFF
TARGET	OFF

TABLE V. Radar transmit frequencies.

TRANSMIT FREQUENCY (GHz)	STALO FREQUENCY (GHz)	CHANNEL NO.
9.205	9.145	1
9.213	9.153	2
9.221	9.161	3
9.229	9.169	4
9.237	9.177	5
9.245	9.185	6

3. On the AN/VPM-2 system tester, set POWER switch to ON. Check that the digital display is illuminated.

4. Observe that the digital display indicates +24.0 $\pm$ 2.0 Vdc.

5. Set the VOLTAGE SELECT switch to the following positions. Note the digital display at each setting.

Voltage Select	Digital Display (Vdc)
VOLT TP	0 $\pm$ 0.015
+5	+5.0 $\pm$ 0.2
+15	+15.0 $\pm$ 0.3
-15	-15.0 $\pm$ 0.3
+28	+28.0 $\pm$ 0.6

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b. Self-test.

1. RF input.

NOTE

Be sure that the pin on the flexible waveguide coupling is properly keyed to the RADAR port. Be sure that both waveguide sections are properly aligned.

(a) Connect flexible cable W5 between the RADAR port and the SELF TEST port of the AN/VPM-2 system tester.

(b) Set the FUNCTION SELECT switch to RF INPUT.

(c) Adjust the RF LEVEL control to obtain a digital display reading of 0.500 ± 0.075 Vdc. The RF LEVEL control setting should be greater than 3.0 (dB).

(d) Set the FUNCTION SELECT switch to RF MONITOR.

(e) Adjust the RF LEVEL control to obtain a digital display reading of +2.50 ± 0.075 Vdc.

(f) Vary the PHASE CONTROL over its entire range. The digital display should remain between +2.0 and +3.0 Vdc. Reset the PHASE CONTROL to midrange.

(g) Adjust the RF LEVEL control to obtain a digital display reading of +2.50 ± 0.075 Vdc.

(h) Set the RANGE SELECT switch to R1, R2 and R4. The digital display should remain between +1.30 and +3.70 Vdc.

(i) Set the RANGE SELECT switch to R3. Set the CLUTTER switch to ON. Set the CLUTTER LEVEL control to 20 dB.

(j) Adjust the RF LEVEL control for a digital display reading of 2.50 ± 0.075 Vdc.

NOTE

The 2.50 ± 0.075 Vdc reading is a reference level injected into the target and clutter channels. Monitor and maintain this voltage throughout the procedure unless instructed otherwise.

(k) Set the FUNCTION SELECT switch to RF SELF TEST.

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(l) Adjust the CLUTTER BALANCE for a digital display within ± 0.1 Vdc of the value indicated on the AN/VPM-2 system tester calibration chart for RF self-test.

(m) Set the FUNCTION SELECT switch to RF MONITOR. Repeat step (j) above.

(n) Set the FUNCTION SELECT switch to RF SELF TEST.

(o) Set the CLUTTER LEVEL control to 25 dB.

(p) Adjust the RF LEVEL control to obtain a digital display reading of +0.750 ± 0.250 Vdc.

(q) Set CLUTTER switch to OFF. DVM should indicate less than +0.075 Vdc.

(r) Set the TARGET switch to ON and the TARGET LEVEL control to 27 dB.

(s) Set the TARGET switch to OFF. The digital display should indicate less than +0.075 Vdc.

(t) Set the CLUTTER switch to ON and note the reading on the digital display.

(u) Set the CLUTTER switch to OFF and the TARGET switch to ON.

(v) Adjust the TARGET BALANCE control for the same digital display reading obtained in step (t) above within ± 0.025 Vdc. Set TARGET switch to OFF.

(w) Repeat steps (t) through (v) above until the digital display indications in steps (t) and (v) are within ± 0.025 Vdc. Do not readjust the CLUTTER BALANCE control.

(x) Set the CLUTTER and TARGET switches to OFF.

2. RF LEVEL attenuator.

(a) Set the FUNCTION SELECT to RF MONITOR.

(b) Adjust the RF LEVEL control for a reading of +3.16 ± 0.025 Vdc on the digital display. Record the RF LEVEL control setting.

(c) Increase (higher number on dial) the RF LEVEL setting by 5.0 dB. The digital display should indicate 1.00 ± 0.25 Vdc.

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3. CLUTTER LEVEL attenuator.

(a) Set the FUNCTION SELECT to RF MONITOR.

(b) Set the CLUTTER switch to ON.

(c) Adjust the RF LEVEL control for a reading of 3.0 ± 0.075 Vdc on the digital display. Record the RF LEVEL control setting.

(d) Set the FUNCTION SELECT to RF SELF TEST.

(e) Adjust the CLUTTER LEVEL control for a reading of +3.0 ± 0.075 Vdc on the digital display. Record the CLUTTER LEVEL control setting.

(f) Increase (higher number of dial) the RF LEVEL setting by 5.0 dB.

(g) Decrease (lower number on dial) the CLUTTER LEVEL control to indicate 3.0 ± 0.075 Vdc on the digital display. Record the CLUTTER LEVEL control setting.

(h) Subtract the CLUTTER LEVEL control setting in step (g) above from the CLUTTER LEVEL control setting in step (e) above. The reading should be 5.0 ± 0.6.

(i) Repeat steps (f) through (h) above in 5.0 dB increments. The last increment may be less than 5.0 dB.

(j) Set the CLUTTER switch to OFF.

4. TARGET LEVEL attenuator.

(a) Set the FUNCTION SELECT switch to RF MONITOR. Set the CLUTTER switch to ON.

(b) Adjust the RF LEVEL control for a reading of +3.0 ± 0.075 Vdc on the digital display. Record the RF LEVEL control setting.

(c) Set the FUNCTION SELECT to RF SELF TEST, the CLUTTER LEVEL control to 25 dB, and the TARGET LEVEL control to 27 dB (fully cw).

(d) Increase (higher number on dial) the CLUTTER LEVEL setting by 5 dB. Record the digital display indication.

(e) Set the CLUTTER switch to OFF and the TARGET switch to ON.

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(f) Increase (higher number on dial) the TARGET LEVEL control for the same DVM indication (within ± 0.075 Vdc) obtained in step (d). Subtract 27 (original setting of step (c)) from this setting. The result should be 5.0 ± 0.6.

NOTE

Self-test of the AN/VPM-2 system tester has now been completed for the radar transmitter frequency selected. If another transmitter frequency is selected, a complete self-test of the system tester must be conducted for that frequency.

(g) Disconnect the flexible waveguide from SELF TEST port and replace SELF TEST cover.

4.7.3.7.2 Receiver test.

a. Control settings. Set the AN/VPM-2 radar system tester controls as follows:

Control	Position
FUNCTION SELECT	VOLTAGE
VOLTAGE SELECT	+24
RF LEVEL	20 dB
FREQ SELECT	Radar transmit frequency (see Table V)
MODE SELECT	RADAR
PHASE-CONTROL	Midrange
CLUTTER	OFF
CLUTTER LEVEL	80 dB
RANGE SELECT	R3
TARGET LEVEL	100 dB
TARGET FREEZE	OFF
TARGET	OFF
POWER	ON

b. Waveguide W5 connection.

1. Disconnect the rotary joint of the antenna, Unit 1, by removing 4 screws, 4 flat washers and 4 lock washers from the waveguide section as shown in Figure 6.

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2. Push the waveguide section away from the rotary joint being careful not to destroy or loosen the RF seal located between the rotary joint and waveguide section. Connect the 90 degree waveguide from the accessory kit to W5 and connect W5 to the rotary joint as shown in Figures 6 and 7.

CAUTION

Ensure that waveguide W5 is connected between the radar and AN/VPM-2 system tester for proper termination to prevent damage to equipment.

c. Transmit frequency determination. Open cover of radar transmitter Unit 2 and determine radar transmit frequency by adding 60 MHz to the frequency marked on solid state crystal of X-band source (refer to Table V).

d. Radar turn-on preliminary.

1. Turn on the system POWER switch.

2. Pull the service interlock switch S3 to the maintenance position.

3. Wait a minimum of 132 seconds and then set the SERV RAD CONT switch located on the inside of the top cover of R/T Unit 2 to ON.

4. The digital display should indicate +24.0 $\pm$ 2.0 Vdc.

5. Set the VOLTAGE SELECT to the following positions and observe the digital display.

Voltage Select	Digital Display (Vdc)
+5	+5.0 $\pm$ 0.2
+15	+15.0 $\pm$ 0.3
-15	-15.0 $\pm$ 0.3
+28	+28.0 $\pm$ 0.6

6. Ensure that the LOCK ON indicator is not illuminated and the SIGNAL CHANNEL meter indicates less than 1.0 Vdc.

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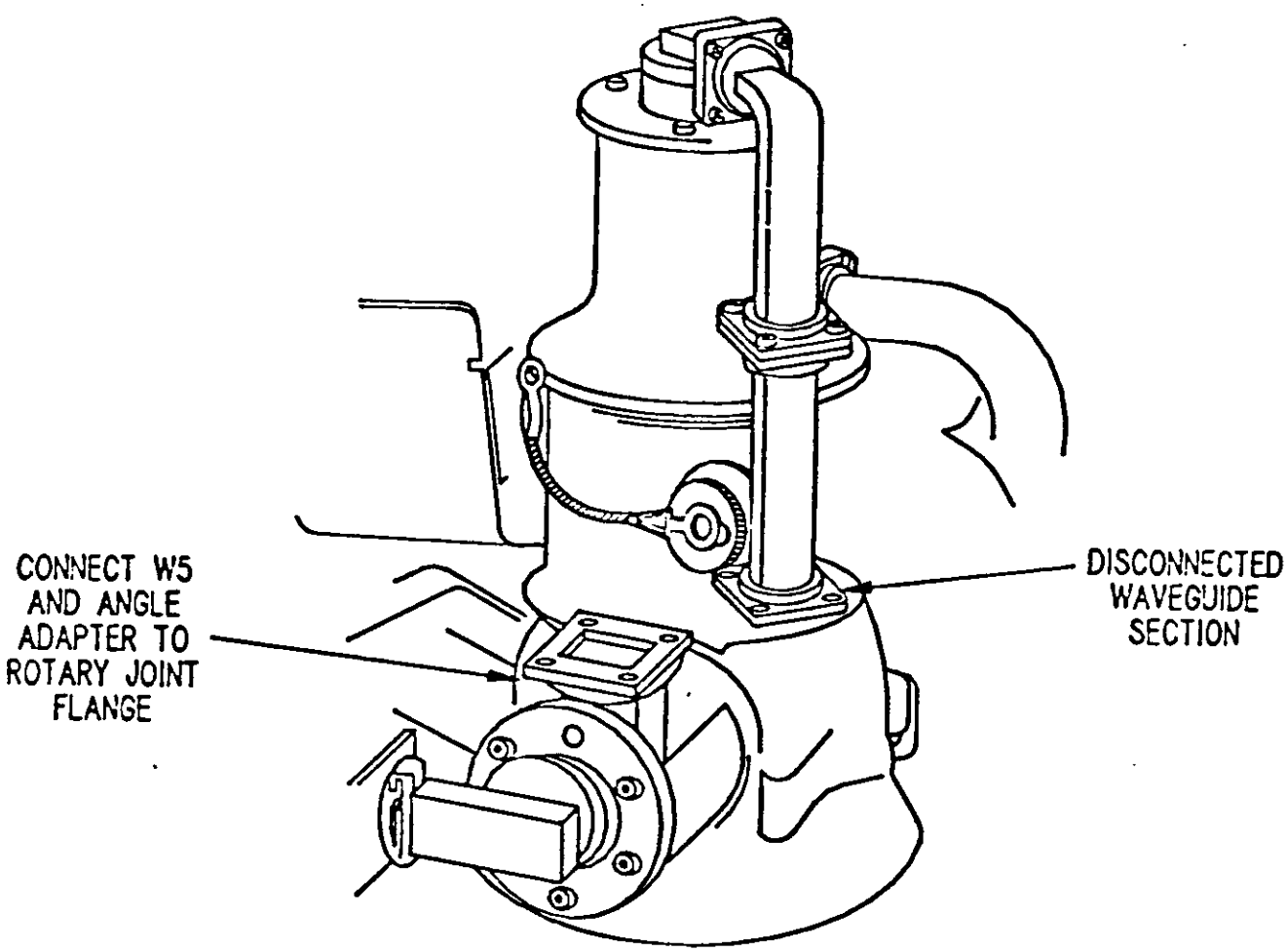


FIGURE 6. Antenna Unit 1, rotary joint disconnection.

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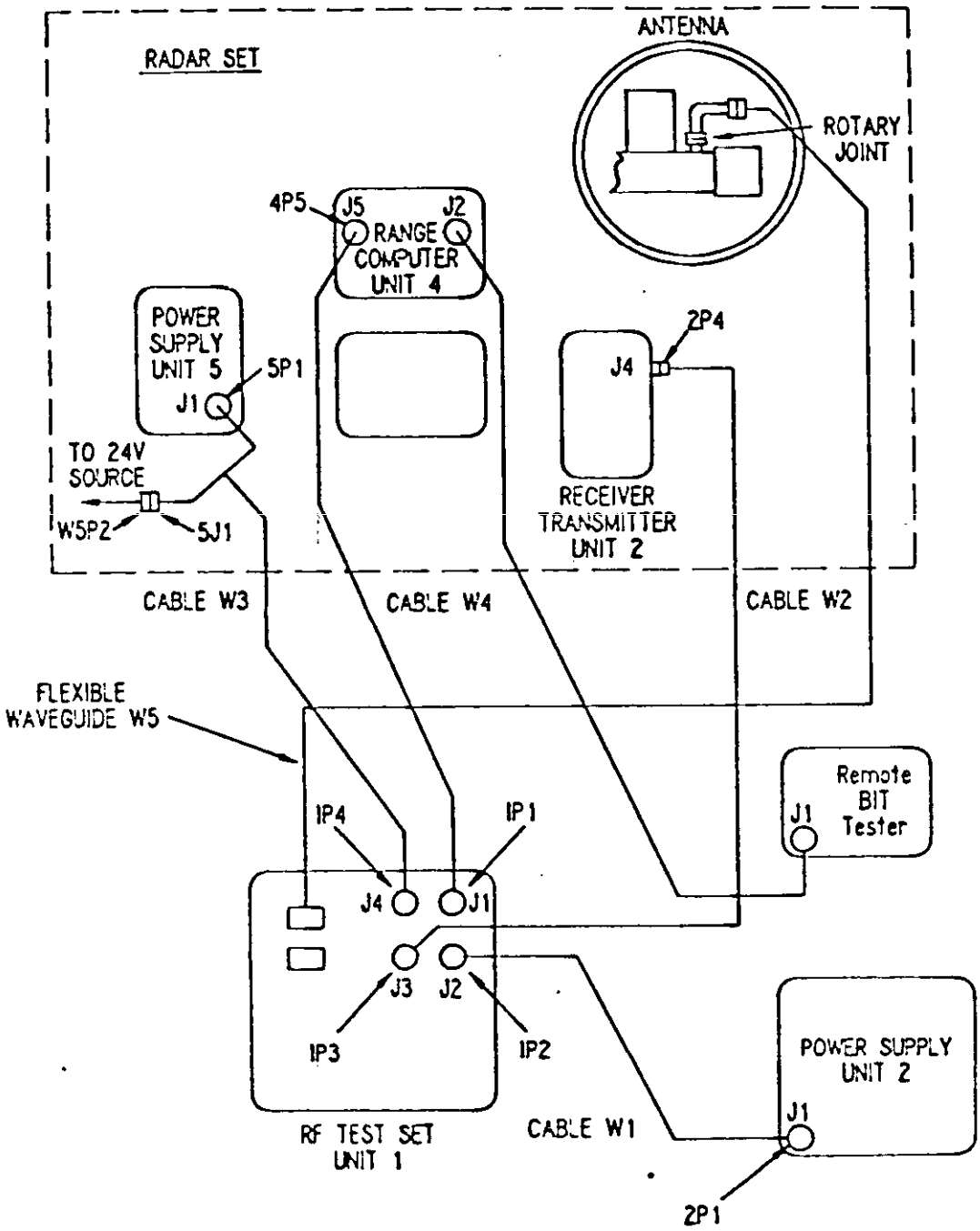


FIGURE 7. AN/VPM-2 system tester, RADAR TEST equipment setup.

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4.7.3.7.3 TARGET LOCK-ON indicator and CLUTTER LOCK-ON. With the radar in the radiate mode and TARGET switch on, the TARGET LEVEL attenuator should be adjusted until the TGT LOCK ON light on (item 29, Table III) lights without flicker (+28 V between 5J2-5(+) and E1(-) on Unit 5). Then the S5 RANGE switch of Unit 4 of the radar is put in the FREEZE position, the AN/VPM-2 TARGET switch is turned OFF, and the CLUTTER switch ON. Set CLUTTER LEVEL attenuator to 80 dB and CLUTTER LOCK-ON switch on Unit 2 to test position. With headset (item 26, Table III) connected to the DOPPLER AUDIO test jack on Unit 2 of the radar, there must be a doppler audio signal clearly audible in the headset.

4.7.3.7.4 Range and mode data.

a. Set the FUNCTION SELECT switch to RF MONITOR and the RANGE SELECT switch to R1.

b. Adjust the RF LEVEL control to obtain a reading of +2.500 ± 0.075 Vdc on the digital display.

c. Set the TARGET switch to ON.

d. Adjust the TARGET LEVEL control until the LOCK ON indicator illuminates.

e. Decrease (lower number on dial) the TARGET LEVEL control setting by 10 dB.

f. Set the FUNCTION SELECT switch to RANGE.

g. Set RBT for range measurement (refer to Table VI).

TABLE VI. Range and range rate.

RBT LED BIT WEIGHTS FOR RANGE AND RANGE RATE

(Range and range rate values are equal to the sum of the bit weights of the illuminated LEDs.)

LED	RANGE	RANGE RATE
DS13	2560 meters	---
DS14	1280	160 m/sec
DS15	640	80
DS16	320	40
DS01	160	20
DS02	80	10
DS03	40	5
DS04	20	2.5
DS05	10	1.25
DS06	5	0.625
DS07	2.5	---
DS08	1.25	---

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TABLE VI. Range and range rate. (continued)

FOR RANGE MEASUREMENTS (4.7.3.7.4)		FOR RANGE RATE MEASUREMENTS (4.7.3.7.11)	
Set the REMOTE BIT TESTER switches to the following positions.		Set REMOTE BIT TESTER switches to positions shown below.	
Switch	Position	Switch	Position
STATION 1234	0 0 0 0	STATION 1234	10 0 10 0
SEND	UP	SEND	UP
UNIT 4 TEST/ NORMAL	NORMAL	UNIT 4 TEST/ NORMAL	UNIT 4 TEST
Depress and release Reset/Lamp-test pushbutton.		Depress and release Reset/Lamp-test pushbutton.	

h. Adjust the PHASE CONTROL for maximum stable indication on RBT LEDs (refer to Table VI).

i. Read and record the range indicated on the RBT. The range shall be equal to the value listed on AN/VPM-2 calibration sticker ± 10 meters for ranges of 1,000 meters or less and ± 1% for ranges greater than 1,000 meters.

NOTE

Allow 3 seconds for the radar to reacquire the target signal each time the RANGE SELECT switch is changed.

j. Set the RANGE SELECT switch to R2, R3 and R4 and repeat steps h and i above.

4.7.3.7.5 SCV without AGC. First perform the Minimum Discernible Signal (MDS) test (a) and then the Sub-Clutter Visibility (SCV) measurement procedure (b). In both cases, operate the RADIATE switch to RADIATE. Preset the CLUTTER LEVEL dial to 80 dB and the CLUTTER switch to OFF.

a. Minimum Discernible Signal (MDS).

1. Set the TARGET switch to ON and the FUNCTION SELECT to RF MONITOR.

2. Adjust the RF LEVEL control and maintain a digital display indication of +2.5 ± 0.075 Vdc for the remainder of this procedural step (a).

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3. Decrease (lower number on dial) the TARGET LEVEL control until the LOCK ON indicator illuminates. Note the TARGET LEVEL dial indication.

4. Increase (higher number on dial) the TARGET LEVEL by 10.0 dB. The LOCK ON indicator should not illuminate after an approximate 3 second delay.

NOTE

The calculated MDS acceptability criteria is 80-88 dBm.

5. Repeat steps 3 and 4 above until the last reading obtained in step 3 above is within 1.0 dB of the previous reading. Record this last reading. This is the radar system MDS.

6. Set the TARGET switch to OFF and the TARGET LEVEL control to 100 dB.

b. Sub-Clutter Visibility (SCV) Measurement Procedure.

1. Set the RANGE SELECT switch to R4.

2. Connect the TPS shorting plug (P/O accessory kit) to J6 of receiver, Unit 3.

3. Set the FUNCTION SELECT switch to RF MONITOR.

4. Set the CLUTTER switch to ON.

5. Check the RF LEVEL control adjustment and maintain a reading of 2.500 $\pm$ 0.075 Vdc on the digital display for the remainder of this procedural step b.

6. Adjust the CLUTTER LEVEL control until the LOCK ON indicator is illuminated. Note the CLUTTER LEVEL dial reading. (Light does not have to stay locked on.)

7. Increase (higher number on dial) the CLUTTER LEVEL control setting by 10.0 dB. The LOCK ON indicator should not be illuminated after an approximate 3 second delay.

8. Carefully repeat steps 6 and 7 above until a reading obtained in step 6 above is within 1.0 dB of the previous reading.

9. Record the CLUTTER LEVEL dial reading obtained in step 8 above. This is the lock on clutter power (no AGC).

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10. Determine the SCV by subtracting the lock on clutter power obtained in step 9 (no AGC) from the minimum discernible signal (MDS) recorded in step a.5. above. The SCV acceptability criteria is 40 dB minimum.

11. Repeat b.1.-b.10. for the remaining three target ranges.

12. Repeat step b.1.-b.11. for the remaining five channels of the radar.

13. Set the CLUTTER switch to OFF, the CLUTTER LEVEL control to 80 dB, and the RANGE SELECT switch to R1. Remove the TPS shorting plug from J6 of receiver, Unit 3, and return it to the accessory kit.

4.7.3.7.6 AGC characteristics.

a. Linearity.

1. Set the CLUTTER LEVEL control to 80 dB and the TARGET LEVEL control to 100 dB.

2. Set the CLUTTER switch to ON.

3. Set the FUNCTION SELECT switch to RF MONITOR. Adjust the RF LEVEL control and maintain a digital display indication of 2.500 $\pm$ 0.075 Vdc for the remainder of this linearity check.

4. Decrease the CLUTTER LEVEL control setting slowly to 5 dB while observing that the LOCK ON indicator does not illuminate.

5. Set the CLUTTER switch to OFF and the TARGET switch to ON.

6. Decrease (lower number on dial) the TARGET LEVEL control until the LOCK ON indicator illuminates. (Note: This is the MDS level.)

7. Set the CLUTTER switch to ON. The LOCK ON indicator should not be illuminated after approximately a 3 second delay.

8. Increase (higher number on dial) the CLUTTER LEVEL control setting until the LOCK ON indicator is illuminated.

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9. Decrease (lower number on dial) the CLUTTER LEVEL setting by 10.0 dB. The LOCK ON indicator should not be illuminated after approximately a 3 second delay.

10. Increase (higher number on dial) the CLUTTER LEVEL control until the LOCK ON indicator is illuminated. Note the CLUTTER LEVEL dial setting at the lock-on point.

11. Carefully repeat steps 9 and 10 above until the readings obtained in step 10 are within 1.0 dB.

12. Record the CLUTTER LEVEL dial setting obtained in step 10 above. This is the LOCK ON CLUTTER POWER (with AGC).

13. Determine the AGC cut-in points by subtracting the LOCK ON CLUTTER POWER recorded in step 12 from the Minimum Discernible Signal (MDS) recorded in step 4.7.3.7.5.a.5. The AGC cut-in acceptability criteria is 38 to 40 dB.

14. Decrease (lower number on dial) the CLUTTER LEVEL control setting for three 10 dB decrements noting the three correspondingly decreased TARGET LEVEL settings required to re-establish LOCK ON each time. The allowable TARGET LEVEL tolerances are as follows:

CLUTTER LEVEL Dial		TARGET LEVEL Dial
*L.O.C.P.-10 dB --	results in --	MDS - (9 to 14 dBm)
present clutter -- setting-10 dB	results in --	present target setting- (9 to 16 dBm)
present clutter -- setting-10 dB	results in --	present target setting- (9 to 18 dBm)

*L.O.C.P. means LOCK ON CLUTTER POWER (with AGC).

15. Set the TARGET and CLUTTER channel switches to OFF.

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4.7.3.7.7 SCV with AGC.

a. Set the TARGET LEVEL control to 88 dB; set the CLUTTER LEVEL control to 40 dB. Place the radiate switch in the radiate position. If the radar does not lock onto the target, decrease (lower number on dial) the TARGET LEVEL control in 1 dB steps until three successive solid locks can be achieved. The TARGET LEVEL attenuation shall exceed the CLUTTER LEVEL attenuation by not less than 40 dB.

b. Repeat step a with initial setting of TARGET LEVEL at 57 dB and CLUTTER LEVEL at 15 dB. The TARGET LEVEL attenuation shall exceed the CLUTTER LEVEL attenuation by not less than 40 dB.

4.7.3.7.8 Active ECM indication.

a. Turn OFF the CLUTTER switch and adjust the TARGET LEVEL control until the radar just acquires the target. Repeat for three successive locks and note the TARGET LEVEL control setting.

b. With the radar locked onto the target, decrease (lower number on dial) the TARGET LEVEL control until the active ECM lamp on the system control unit illuminates. The TARGET LEVEL control setting shall be not less than 15 dB and not more than 25 dB lower than the TARGET LEVEL control setting noted in step a.

c. Set the TARGET switch to OFF.

d. Take the radar out of the radiate condition.

4.7.3.7.9 Three-second fade logic. With the test equipment set up as shown in Figures 3, 7 and 8, and with the radar in the standby mode, program the simulator as follows:

START	(FT)	1,000
STOP	(FT)	11,000
VELOCITY		INC. 100 FT/S
RANGE		INCREASING

Depress the RESET button of the simulator. Connect the LEFT and RIGHT channels of the recorder to Unit 4. Connect the LEFT channel to 4TP15 and the RIGHT channel to 4TP13 (returns to Unit 4 chassis). With the deflection sensitivity of both channels set to 5 V/cm and the speed to 20 mm/s, turn on the recorder and press the START button on the simulator. On Unit 4, press switch S6 RATE to RESET and hold and simultaneously press the FADE button on the simulator and hold both buttons for 5 seconds and then release both. The elapsed time between where RIGHT channel deflects and LEFT channel deflects must be 3.00 $\pm$ 0.25 s.

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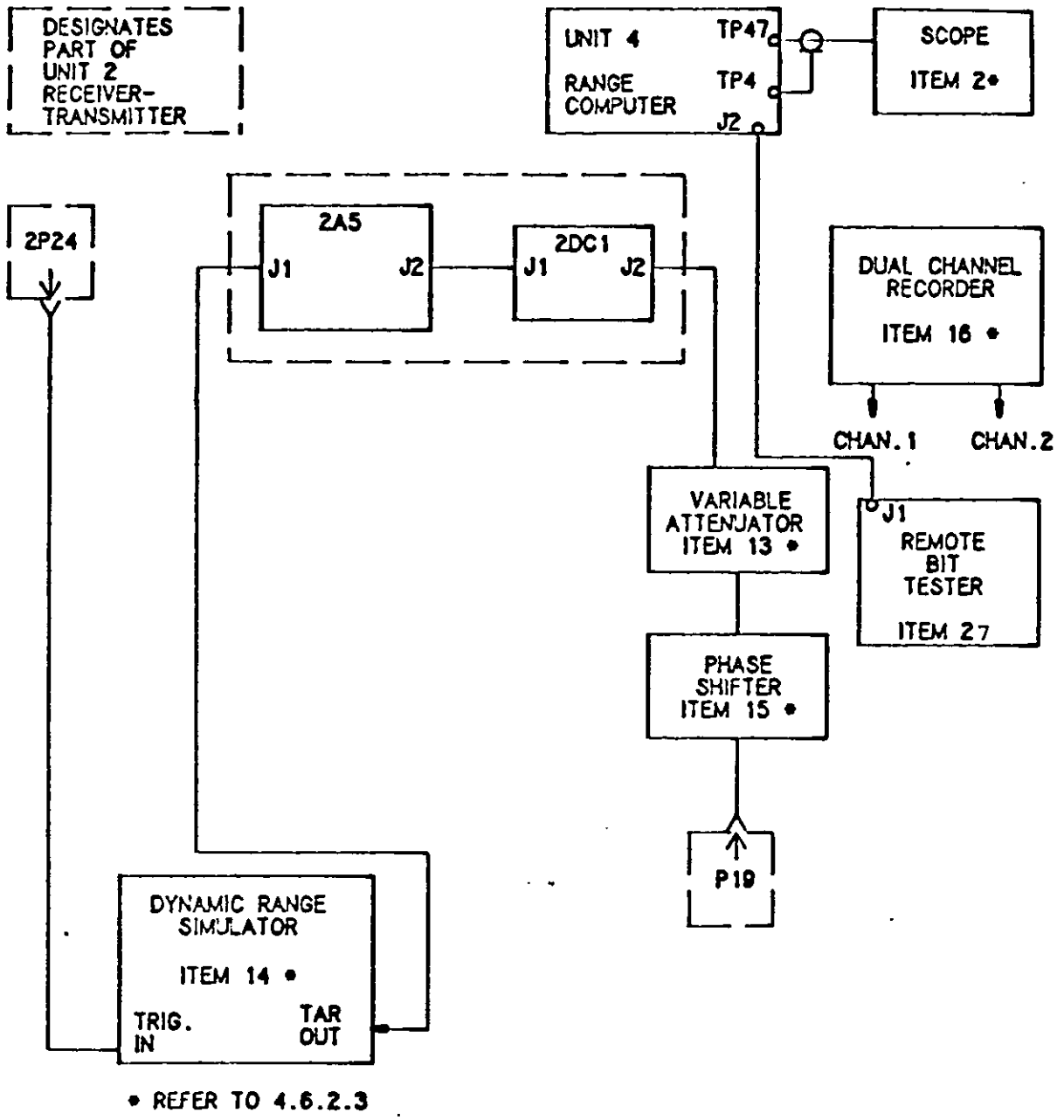


FIGURE 8. Range and range mode test setup.

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4.7.3.7.10 One-second fade logic. The test equipment setup is the same as in 4.7.3.7.9. The recorder speed shall be 20 mm/s during the runs. With reference to Table VII (One-second fade logic tests) program the simulator as specified in RUN 1. Press the RESET button on the simulator, then the START button. Start the recorder, then hold the FADE button on the simulator. The length of time that the radar remains in coast (the time between the two deflections) must be 1.0 ± 0.02 s.

TABLE VII. One-second fade logic tests.

RUN	START (FT)	RANGE	STOP (FT)	VELOCITY (FT/s)
1	1,100	Increase	11,000	00328.1
2	1,100	Increase	11,000	00656.2
3	1,100	Increase	11,000	00984.3
4	11,000	Decrease	1,100	00984.3
5	11,000	Decrease	1,100	00656.2
6	11,000	Decrease	1,100	00328.1

4.7.3.7.11 Range rate lag. The test setup is the same as 4.7.3.7.10. Set the dynamic range simulator as follows:

START	(FT)	12,000
STOP	(FT)	10,782
ACCELERATION		270 FT/s^2
VELOCITY		0 FT/s
RANGE		DECREASE

Set RBT for Range Rate measurement (refer to Table VI).

Adjust the attenuator (item 13, Table III) for 600 mV peak-to-peak; adjust the phase shifter (item 15, Table III) for maximum signal; readjust the attenuator for 600 mV peak-to-peak. Turn on the recorder, allow the target to be acquired by the radar, then depress the START button on the simulator. At the end of the run the range rate must be a minimum of 217 m/s. Set the AN/VPM-2 POWER switch to OFF. Disconnect the AN/VPM-2 from the radar.

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4.7.3.7.12 Minimum range tracking. Set up the test equipment per Figures 3 and 8. Place radar in standby mode. Set RBT for Range measurement (refer to Table VI). Set target simulator (item 14, Table III) for 0 FT/s velocity, START FT at 000999 FT, and zero acceleration. Press the RESET button on the simulator. Decrease the range setting until the RBT reads 250 meters or less. The phase shifter (item 15, Table III) may be adjusted if necessary to achieve the reading.

4.7.3.7.13 Receiver bandwidth. Set up the test equipment as shown in Figure 3. The radar is to be in the standby mode. With the RF signal generator (item 21, Table III) set for 61.0 MHz, the output attenuator set for -30 dBm, and the output terminated in 50 ohms, connect the output to 3J1-D of the radar. With a 50 ohm termination on 3A1J2 (i-f amplifier assembly 10548164) of the radar and the RF voltmeter (item 22, Table III) terminated in 50 ohms connected to 3A1J3, tune the RF signal generator for maximum indication on the voltmeter. Adjusting the frequency of the generator higher than 61 MHz, note the frequency at which the voltmeter reads 3 dB lower than the maximum just obtained. Adjust the frequency of the generator below 61 MHz and note the frequency at which the voltmeter reads 3 dB lower than maximum. The difference between the two frequencies shall be 1.9 $\pm$ 0.2 MHz. On system control unit (item 25, Table III) turn off PRIMARY POWER switch S1.

4.7.3.7.14 Remote BIT assemblies. The Units 2, 4 and 5 remote BIT assemblies shall have demonstrated compliance with MIL-R-70561, MIL-C-70562, MIL-P-70563, respectively.

4.7.3.7.15 Antenna, Unit 1. The antenna shall have demonstrated compliance with MIL-A-70708.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of Note 2 of Drawing 12950877 as specified in paragraph 3.7.

5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with Special Packaging Instruction PS 9360797. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This radar is to be used to acquire and track moving aircraft targets and supply range information to the fire control processor.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

c. Applicable stock number.

d. First article sample requirements.

e. Packaging requirements, if other than specified in Section 5.

f. Serialization requirements, if applicable.

g. Certificate of Conformance for each lot or shipment of product.

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, Attn: AMSMC-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (key word) listing.

Fire Control System
Product Improved Vulcan Air Defense System (PIVADS)
Self-Propelled M163A2, 20MM
Towed M167A2, 20MM

6.6 Changes from previous issue. The previous issues were for ARDEC internal use only. They were not forwarded to Naval Publication and Forms Center (NPFC) for publication. This issue is initial public release for unlimited distribution.

Custodian:
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Preparing activity:
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-R-70560	04 DECEMBER 1991

3. DOCUMENT TITLE
RADAR SET AN/VPS-2A: 9360797 (RANGE ONLY RADAR)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

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c. ADDRESS (Include Zip Code) ATTN: SMACA-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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